



Smart Manufacturing & Automation Laboratory (SMAL) IIT Tirupati



EQUIPMENT CATALOGUE

<https://smalab.org/>

SMART MANUFACTURING AND AUTOMATION LABORATORY (SMAL)



The Smart Manufacturing and Automation Laboratory (SMAL) at IIT Tirupati is a multidisciplinary research facility focused on machine tool dynamics, vibration control, robotics, automation, and intelligent manufacturing systems. The laboratory integrates advanced sensing, data acquisition, computer vision, artificial intelligence, digital manufacturing, and collaborative robotics to develop next-generation smart manufacturing solutions.

SMAL emphasizes both experimental and theoretical research to address challenges in precision manufacturing, industrial automation, condition monitoring, structural dynamics, and human-robot collaboration. The laboratory is equipped with state-of-the-art robotic systems, machine vision technologies, sensor networks, and advanced computational platforms to support cutting-edge research and innovation.

The laboratory actively contributes to the development of technologies for manufacturing, automotive, aerospace, healthcare, and other industrial sectors. Through research projects, industry collaborations, and hands-on training programs, SMAL provides students, researchers, and professionals with practical exposure to emerging technologies and Industry 4.0 applications.

Under the leadership of Govind Narayan Sahu, SMAL is committed to advancing smart and sustainable manufacturing technologies while fostering innovation, skill development, and industry-oriented research.

INTEL REALSENSE D435I DEPTH CAMERA



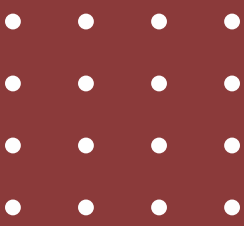
The Intel RealSense D435i combines depth sensing, RGB imaging, and inertial measurement capabilities to support 3D perception, object tracking, navigation, and robotic vision applications. It is widely used in autonomous and intelligent systems.



ZED 2I STEREO CAMERA



The ZED 2i is a professional stereo vision camera capable of high-resolution 3D depth sensing, spatial mapping, and motion tracking. It supports advanced robotics, digital twins, and autonomous navigation applications.



MAGIC LEAP 2 DEVELOPER PRO EDITION



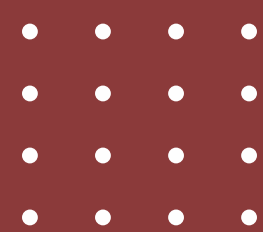
The Magic Leap 2 Developer Pro Edition is an augmented reality headset designed for immersive visualization, human-machine interaction, and mixed reality application development. It enables real-time integration of digital content with physical environments.



BASLER ACE ACA3800-10GM CAMERA



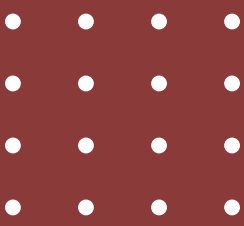
The Basler ace acA3800-10gm is an industrial-grade machine vision camera that provides high-resolution image acquisition for inspection, measurement, and automation systems. It is suitable for demanding computer vision applications.



BASLER LENS C23-0824-5M F8MM



The Basler C23-0824-5M lens is a high-quality machine vision lens optimized for industrial imaging applications. It delivers sharp image quality and accurate optical performance for vision-based inspection systems.



JAKA ZU5 COLLABORATIVE ROBOT WITH FORCE SENSOR AND ONROBOT GRIPPER



The JAKA Zu5 is a collaborative robotic arm with a 5 kg payload and 954 mm reach, designed for automation, assembly, and material handling applications. Integrated with a force/torque sensor, it enables precise force-controlled operations such as assembly, inspection, and human-guided programming. The OnRobot gripper provides adaptive and reliable object handling for pick-and-place and machine-tending tasks. Together, they form a versatile and safe robotic system for advanced industrial and research applications.



SPEED-ADJUSTABLE BELT CONVEYOR (CUSTOMIZED)



A speed-adjustable belt conveyor is used for automated material transportation between workstations in manufacturing, inspection, and robotic handling environments. Adjustable speed control enables flexible operation and synchronization with robotic systems and production processes.



ROBOTIC SAFETY FENCE



A robotic safety fence creates a secure working area around automated systems and industrial robots to protect personnel and equipment. It supports compliance with industrial safety standards while allowing controlled access for maintenance and operation.



TOOLFAST STAINLESS STEEL CLAMPING KIT (34 PIECES)



The Toolfast clamping kit consists of precision-engineered stainless steel clamps and accessories used for securely holding workpieces during machining, assembly, and testing operations. It enhances setup accuracy, repeatability, and operational efficiency.



FIXTURE TABLE WITH BENCH VICE



A fixture table with bench vice provides a stable and versatile workstation for assembly, fabrication, machining, inspection, and prototyping activities. The integrated bench vice securely holds workpieces during cutting, drilling, filing, and mechanical operations.



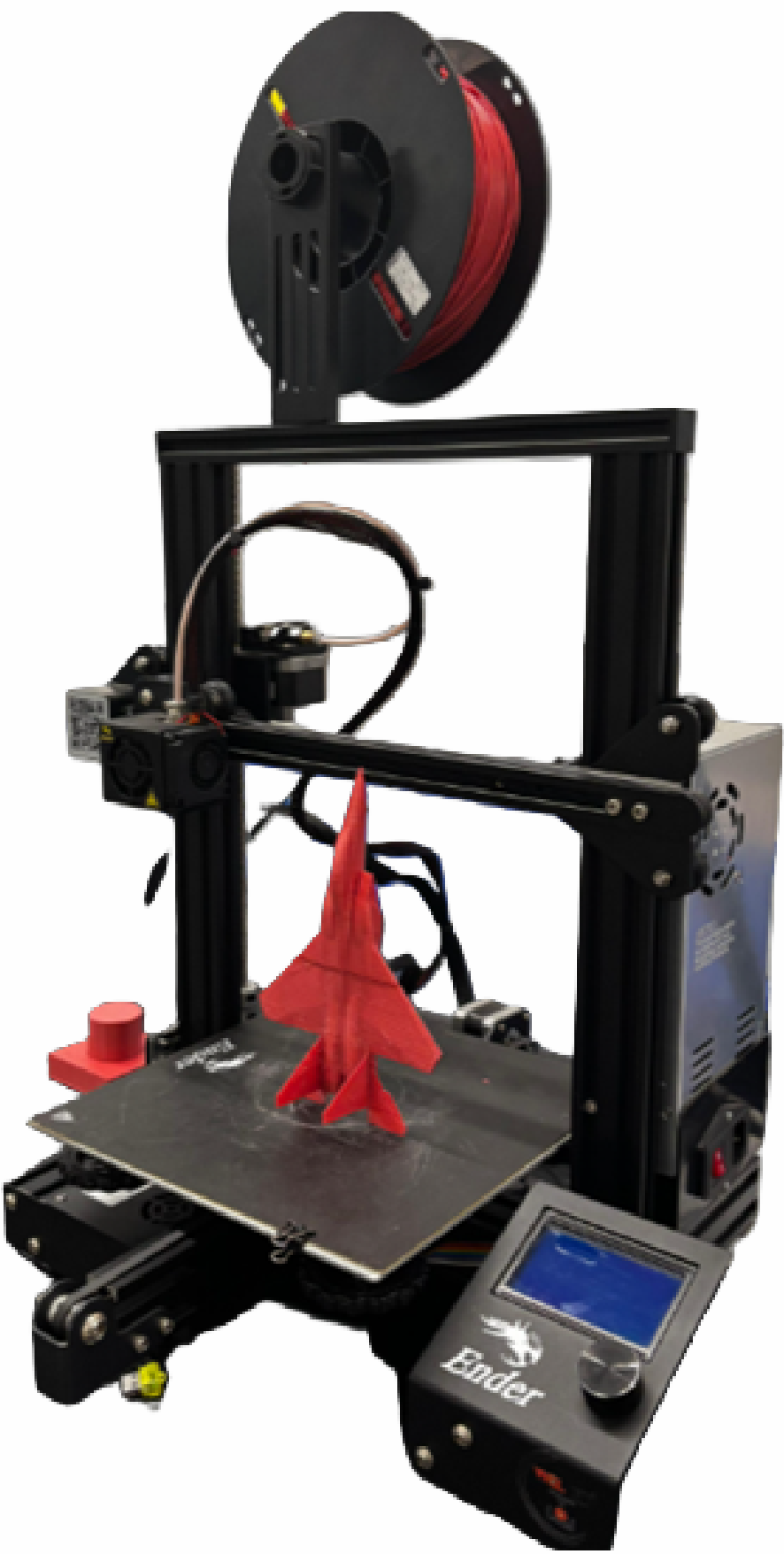
DELL PRECISION T3680 TOWER WORKSTATION



The Dell Precision T3680 workstation features a high-performance Intel Core i9 processor, professional-grade computing capabilities, and fast SSD storage for engineering simulations, CAD design, AI development, and robotics applications. It is optimized for demanding computational workloads and multitasking environments.



CREALITY ENDER 3 DIY 3D PRINTER



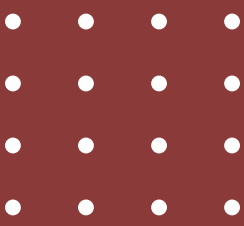
The Creality Ender 3 is an FDM-based 3D printer designed for rapid prototyping, product development, and fabrication of custom mechanical and electronic components. It offers high-precision printing with a large build volume, enabling the production of functional prototypes and engineering models.



TEKTRONIX AFG1022 ARBITRARY/FUNCTION GENERATOR



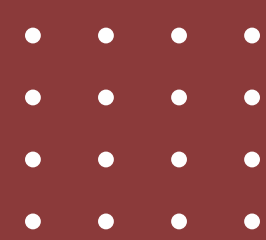
Tektronix AFG1022 Arbitrary/Function Generator is a dual-channel signal generator used to create test waveforms such as sine, square, pulse, ramp, noise, and arbitrary waveforms for circuit design and validation. It offers a 25 MHz maximum output frequency, 14-bit vertical resolution, and a built-in 200 MHz frequency counter for accurate signal generation and measurement. It is widely used in electronics laboratories, communication experiments, sensor simulation, and functional testing.



EKTRONIX TBS1000C SERIES DIGITAL OSCILLOSCOPE



Ektronix TBS1000C Series Digital Oscilloscope is a 2-channel digital storage oscilloscope used for analyzing and debugging electronic signals and circuits. It provides up to 200 MHz bandwidth and 1 GS/s sampling rate, with features like automatic measurements, FFT analysis, and advanced triggering for accurate waveform observation.



NI CDAQ 9183 (COMPACTDAQ)



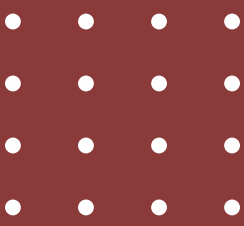
NI cDAQ 9183 is a modular data acquisition system developed by National Instruments (NI) for real-time measurement, sensor interfacing, and industrial testing applications. It supports interchangeable I/O modules for acquiring signals such as voltage, temperature, vibration, strain, and current with high accuracy and synchronization. NI cDAQ systems are widely used in automation, embedded testing, aerospace, and research laboratories.



NI 9201, ANALOG INPUT MODULE



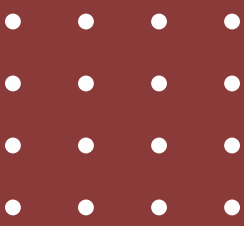
NI 9201 is an 8-channel analog voltage input module for NI CompactDAQ (cDAQ) and CompactRIO (cRIO) systems. It provides $\pm 10\text{ V}$ input range, 12-bit resolution, and up to 500 kS/s aggregate sampling rate, making it suitable for high-speed data acquisition from sensors, test equipment, and electronic circuits. The module also features 100 V overvoltage protection and is widely used in industrial monitoring, automation, and research applications.



NI 9234, IEPE SUPPORTED ANALOG INPUT MODULE



NI 9234 is a 4-channel dynamic signal acquisition module designed for vibration, sound, and condition monitoring applications. It supports IEPE accelerometers and microphones, provides 24-bit resolution, and offers simultaneous sampling up to 51.2 kS/s per channel with built-in anti-aliasing filters. The module is compatible with NI CompactDAQ and CompactRIO systems and supports TEDS smart sensors for automatic sensor identification.



NI USB-6003 MULTIFUNCTION DAQ DEVICE



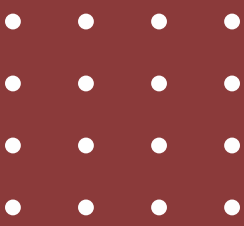
The NI USB-6003 is a 16-bit multifunction data acquisition device designed for precise measurement, monitoring, and control applications. It provides analog and digital I/O interfaces for connecting sensors, actuators, and experimental systems. Its USB connectivity enables real-time data acquisition and analysis in research, robotics, and automation environments.



PASCO WIRELESS 3-AXIS ACCELERATION/ALTIMETER (PS-3223)



PASCO Wireless 3-Axis Acceleration/Altimeter (PS-3223) is a wireless sensor that measures 3-axis acceleration, 3-axis angular velocity, and altitude in real time. It supports acceleration ranges up to +400 g, sampling rates up to 5 kHz, and communicates via Bluetooth Low Energy (BLE) for data logging and analysis using PASCO SPARKvue or Capstone software. It is widely used for motion, vibration, impact, and physics experiments.



IMPACT HAMMER

(MODEL: 086D20, MAKE: PCB
PIEZOTRONICS)



The instrumented impulse hammer is designed for modal testing, vibration analysis, and dynamic characterization of mechanical structures. It features an integrated force sensor with a measurement range of up to 5,000 lbf and interchangeable impact tips for varying excitation conditions. The hammer enables accurate measurement of input excitation forces for Frequency Response Function (FRF) and modal parameter estimation in research and engineering applications.



TRI-AXIAL ACCELEROMETER (MODEL: 356A09, MAKE: PCB PIEZOTRONICS)



The PCB Piezotronics 356A09 is a high-sensitivity tri-axial accelerometer designed for precise measurement of vibrations and dynamic responses in three orthogonal directions. It enables comprehensive vibration monitoring, modal analysis, and structural health assessment of mechanical systems. Its compact design and high measurement accuracy make it suitable for research, testing, and industrial condition-monitoring applications.



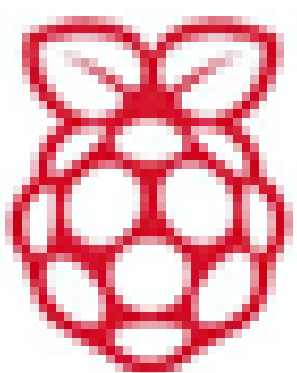
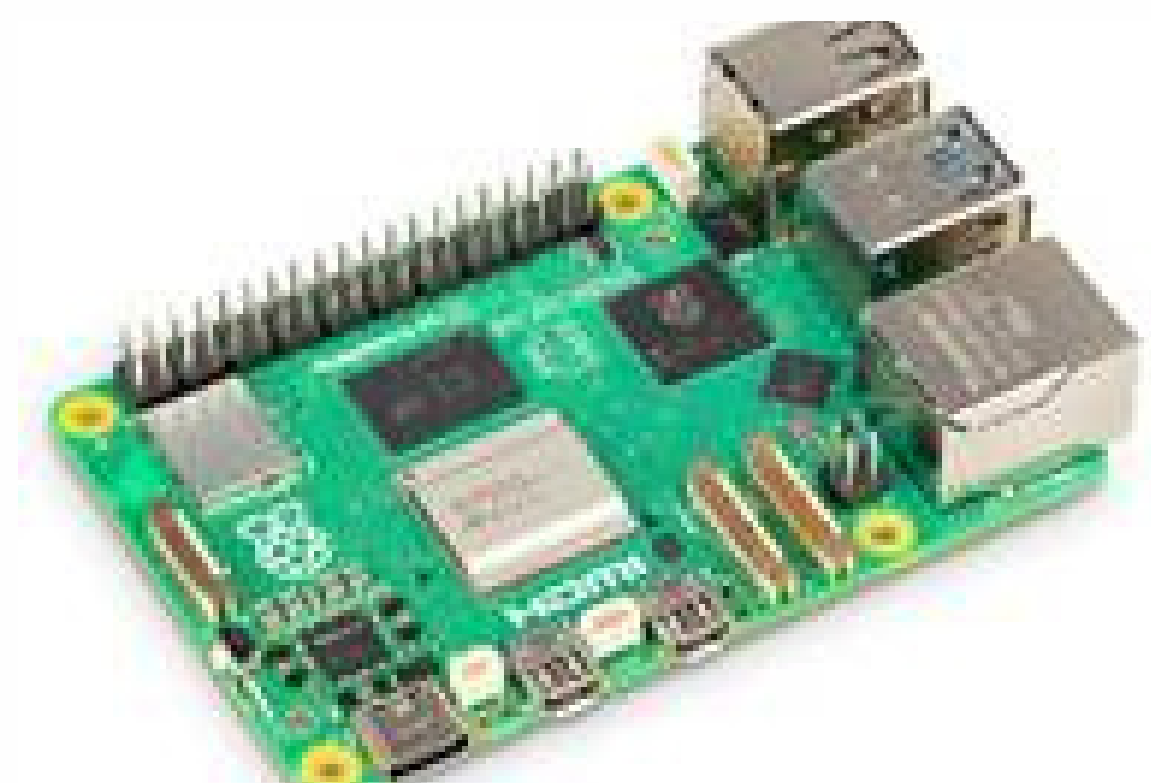
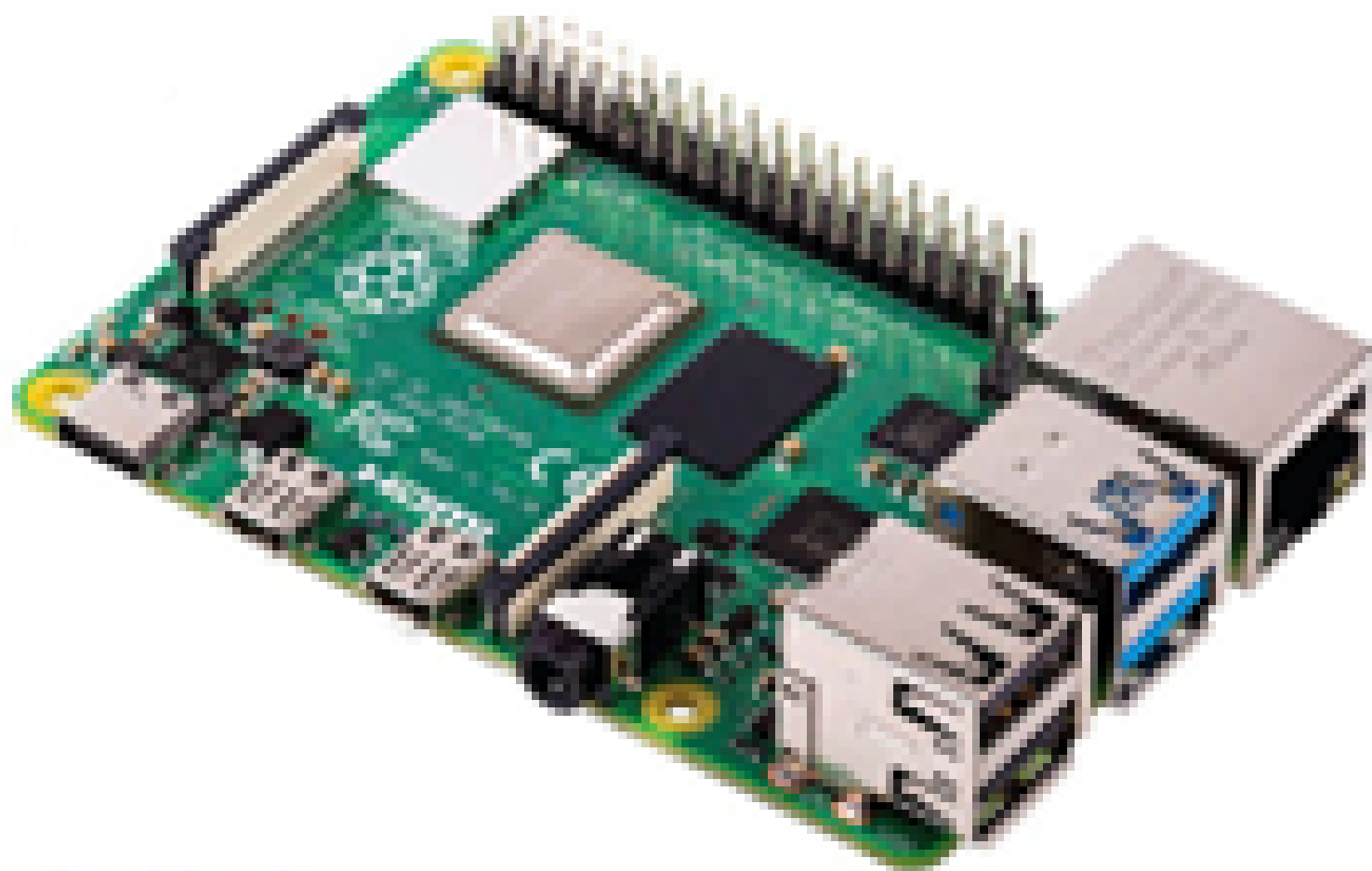
NVIDIA JETSON ORIN NANO SUPER DEVELOPER KIT WITH 1TB SSD



The NVIDIA Jetson Orin Nano Super Developer Kit delivers high-performance AI computing for robotics, computer vision, autonomous systems, and edge AI applications. It enables deployment of advanced machine learning and deep learning workloads on embedded platforms.



RASPBERRY PI 4 & 5



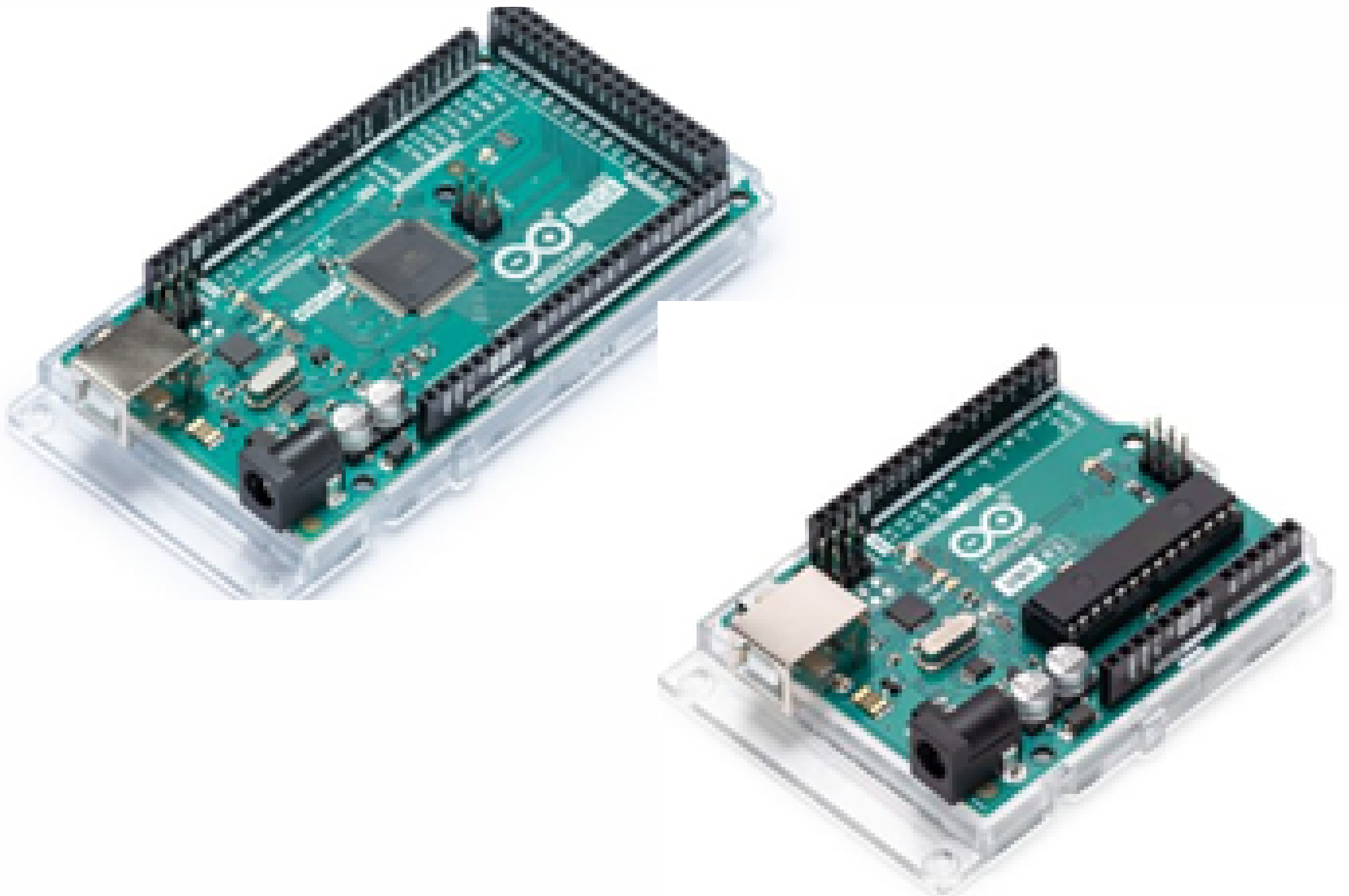
4^B



The Raspberry Pi 4 and Raspberry Pi 5 are compact, high-performance single-board computers designed for embedded systems, IoT, AI, robotics, and edge computing applications. They feature 64-bit quad-core processors, up to 8GB RAM, dual-band Wi-Fi, Bluetooth, Gigabit Ethernet, and multiple connectivity options for advanced development and automation projects.



ARDUINO MEGA 2560 & UNO REV3



The Arduino Mega 2560 and Arduino Uno Rev3 are versatile microcontroller development boards used for embedded systems, robotics, automation, IoT, and prototyping applications. They provide digital and analog I/O interfaces for sensor and actuator integration, enabling rapid development and testing of electronic and control systems.



G1200 DIGITA ELECTRONIC MICROSCOPE



The G1200 digital microscope features up to 1200× magnification and a built-in LCD display for inspecting electronic components, PCBs, and precision assemblies. It assists in soldering, debugging, and quality inspection tasks.

